10/21/2026: Functions

CSCI 246: Discrete Structures

Textbook reference: Sec 24, Scheinerman

Graded Quiz Pickup

Quizzes are in the front of the room, grouped into four bins (A-G, H-L, M-R, S-Z) by last name. The quizzes are upside down with your last name on the back. Come find yours before, during, or after class. Only turn the quiz over if it's yours.

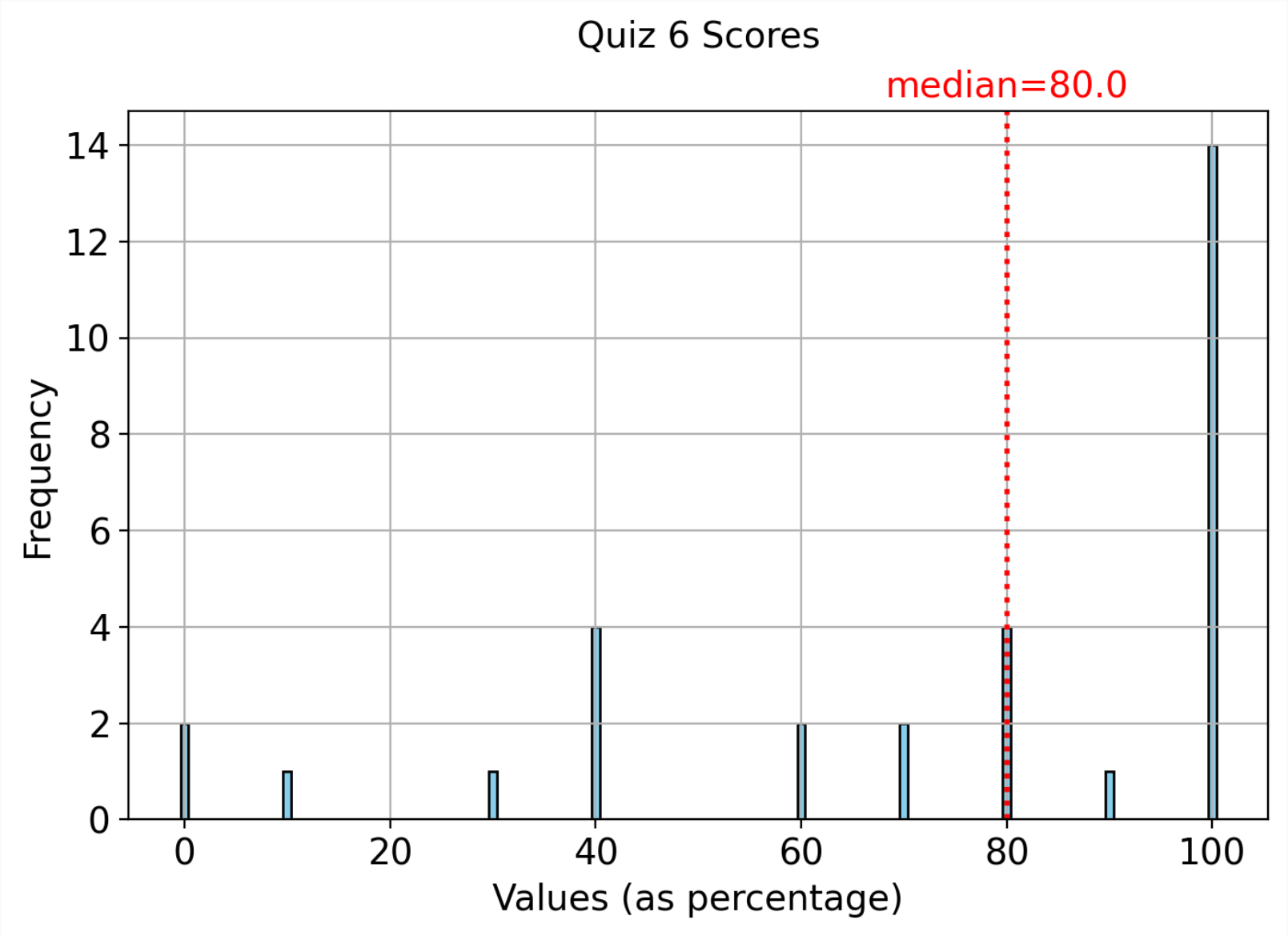
Announcements

- Optional midterm retakes can be scheduled at the testing center. They require at least 2 days advance notice. The midterm retake will re-pull questions from the same study guide. I will write a separate version for each person who decides to retake the exam.
- After today, we will do three classes on computational complexity. The readings for these classes come from the Epp textbook, which can be found in the *Files* section of Canvas.

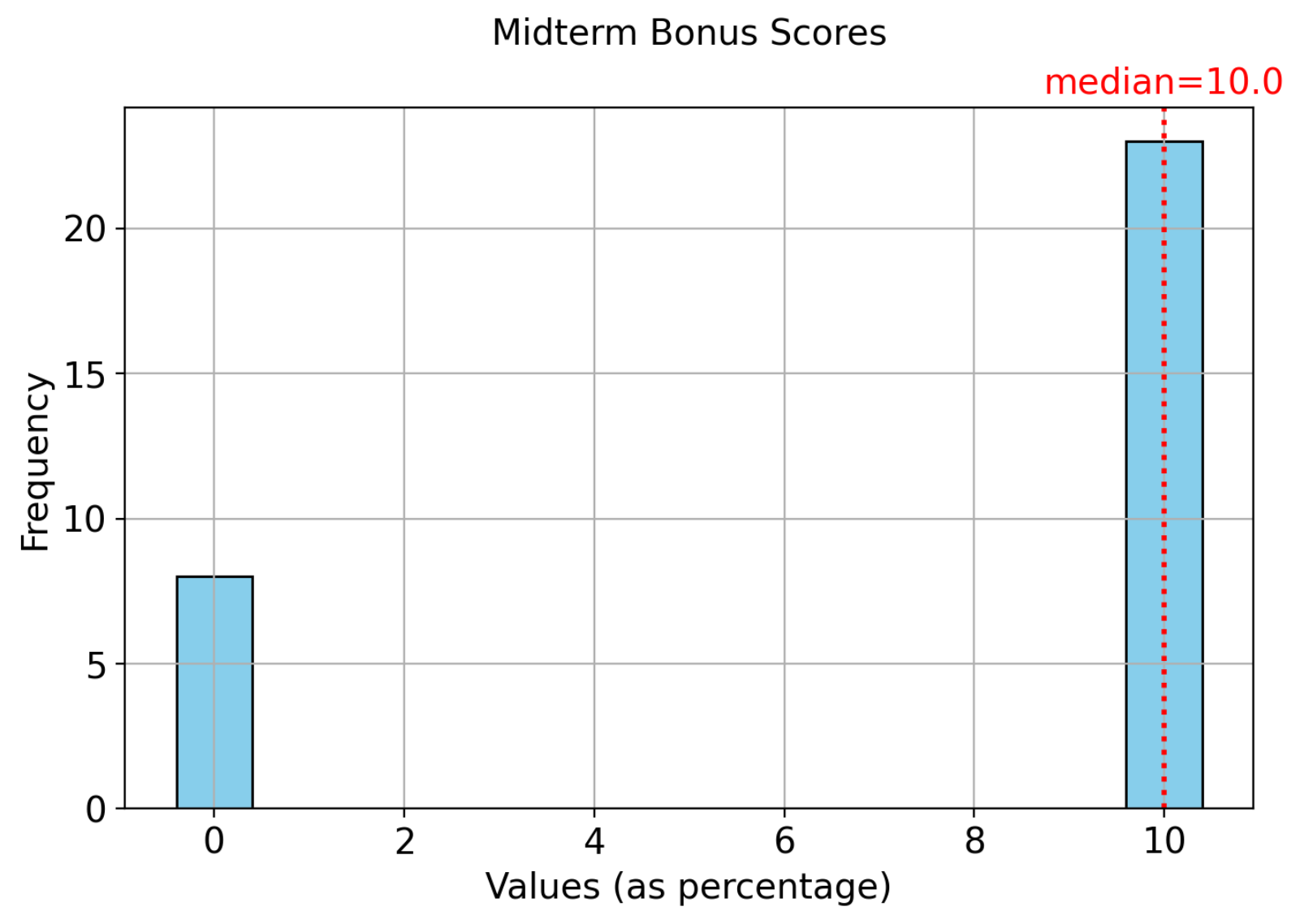
Today's Agenda

- Mini-lecture ($\approx$ 35 mins)
- Group exercises ($\approx$ 15 mins)

Weekly Quiz #6 Scores



Midterm Bonus Scores



Midterm Bonus Scores



Study Guide For Quiz 7

Group exercises. Know how to solve the group exercises from the course meetings on Smallest Counterexample (10/12), Induction (10/14), and Recurrence (10/16).

Readings.

Sec. 21 (Smallest Counterexample)

- Understand Proof Templates 15 and 16 (two versions of the same thing), as used to prove Proposition 21.1.

Sec. 22 (Induction)

- Know how to use Theorem 22.2 / Proof Template 17 to prove Props. 22.3 and 22.5.
- Understand the difference between Proof Template 17 and 18.

Sec. 23 (Recurrence)

- Know how to solve first-order recurrences from Props 23.1 and 23.3. For example, see Example 23.2.
- Know how to solve second-order recurrences from Theorems 23.5 and 23.9. For example, see Example 23.6.
- Know how to use Theorem 23.17 to find a polynomial expression to generate the first few terms of a sequence, as in Example 23.18.

Outline for Mini-Lecture:

- Overview of functions
- Domain, co-domain, and image
- Injectivity and surjectivity
- Inverse functions
- Applications (outside CSCI 246)

Demo

Go the nearest whiteboard and write your name and favorite sport as follows:

Name, Favorite sport.

(Please write your name as your first and last name, or first name and last initial.)

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1. Is $P \subseteq N \times S$?

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1. Is $P \subseteq N \times S$? Is $P \subseteq S \times N$?
2. Is P a relation on $N \times S$?

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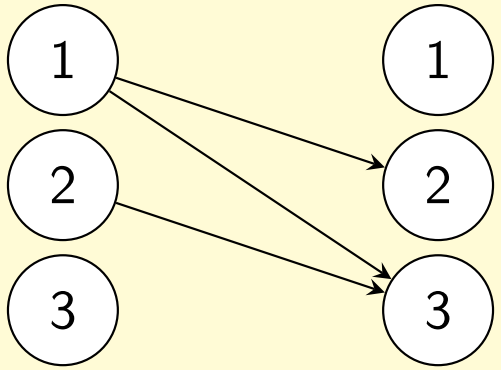
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From relations to functions

Poll

Let $A = \{1, 2, 3\}$. Consider the relation R on A depicted below.

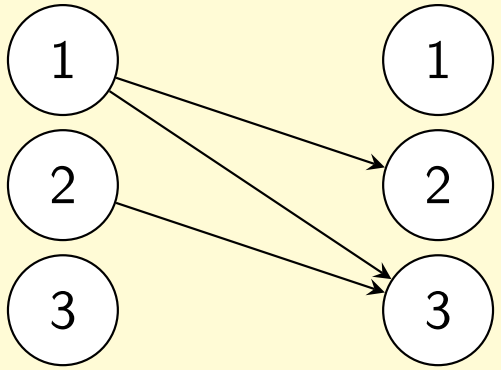


Questions: (1) What is a relation on A in general? (2) How would you describe this relation in words? (3) Is this relation a function?

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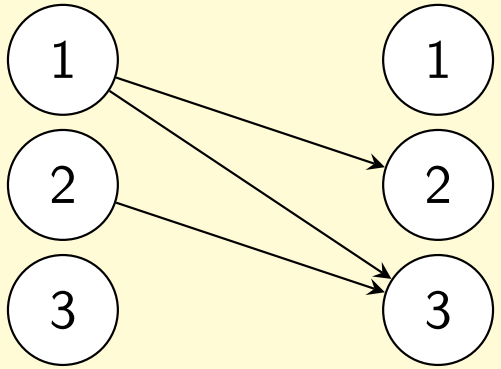
Solutions.

1. A relation on A is a subset of all pairs (x, y) where $x, y \in A$.

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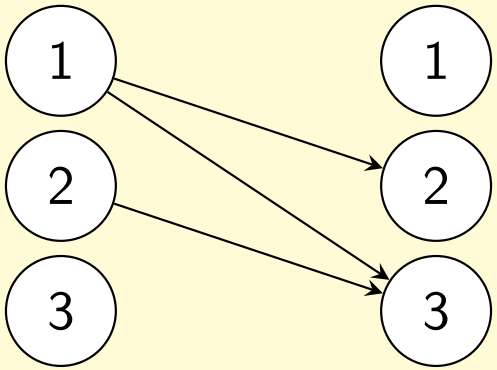
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2. We have $R =$ "$<$". This is the **less than** relation.

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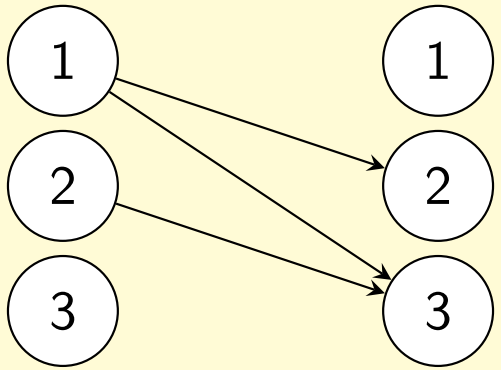
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1. A relation on A is a subset of all pairs (x, y) where $x, y \in A$.
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3. No. A function is like a machine mapping each possible input to an output. But here there are two problems: What are they?

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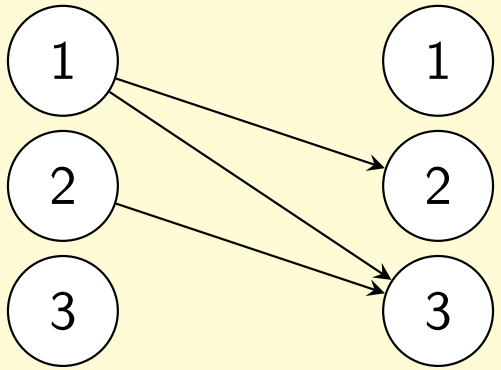
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3. No. A function is like a machine mapping each possible input to an output. But here there are two problems: (a) The input 3 doesn't get mapped to anything. (b) The input 1 gets mapped to 2 things.

Functions: What are they?

Definition (Hampkins pp. 128)

We say that f is **function** from A to B, written $f : A \to B$, if f is a set of pairs $f \subseteq A \times B$ exhibiting the function property : for every $a \in A$, there is a unique $b \in B$ with $(a, b) \in f$.

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Characterization

1. A function is a special type of a relation.
2. We can think of a **function** as a rule that assigns each element of a set A to exactly one element in another set B.

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Notation

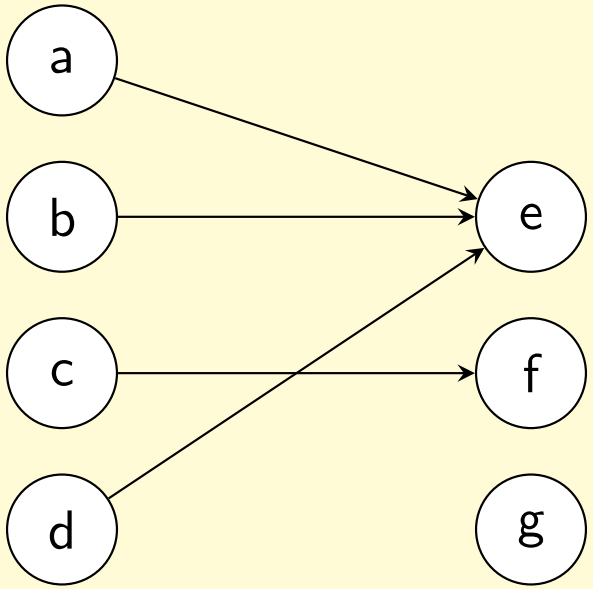
The object b in the definition is denoted by $f(a)$ and is called the value of the function at a.

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We can think of a **function** as a rule that assigns each element of a set A to exactly one element in another set B.

Poll

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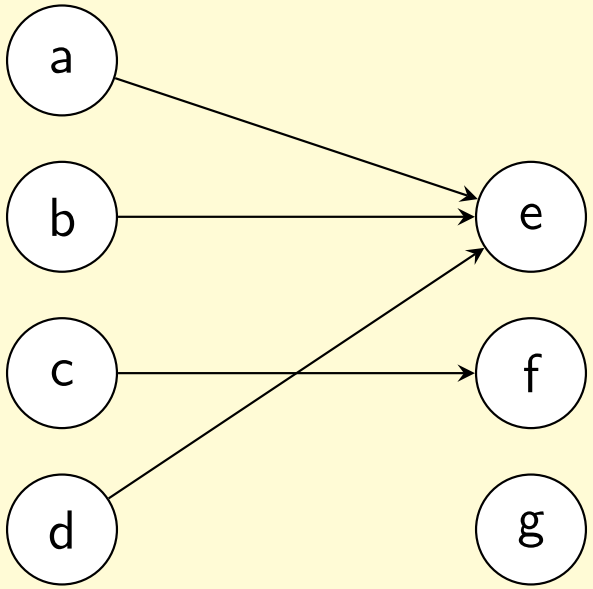
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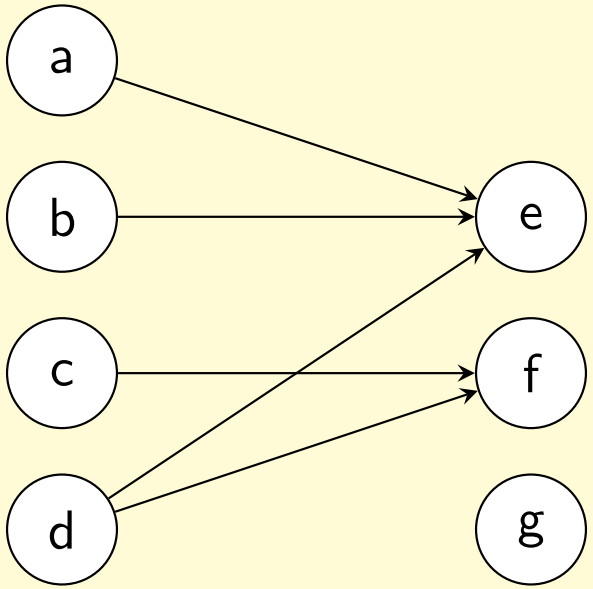
Is R a function? Yes.

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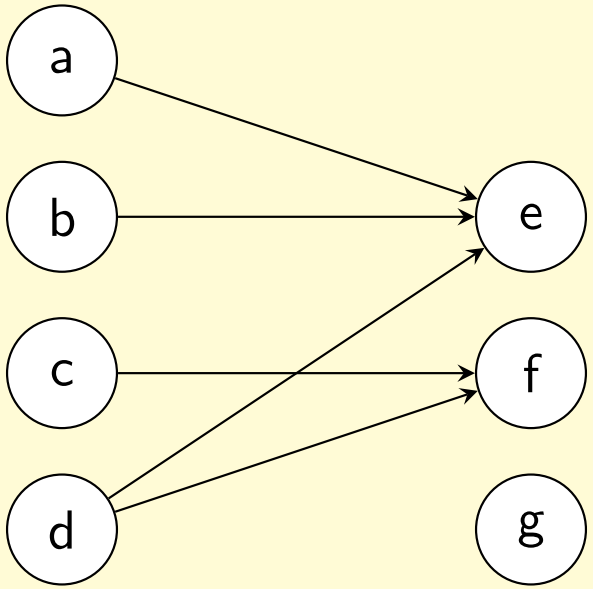
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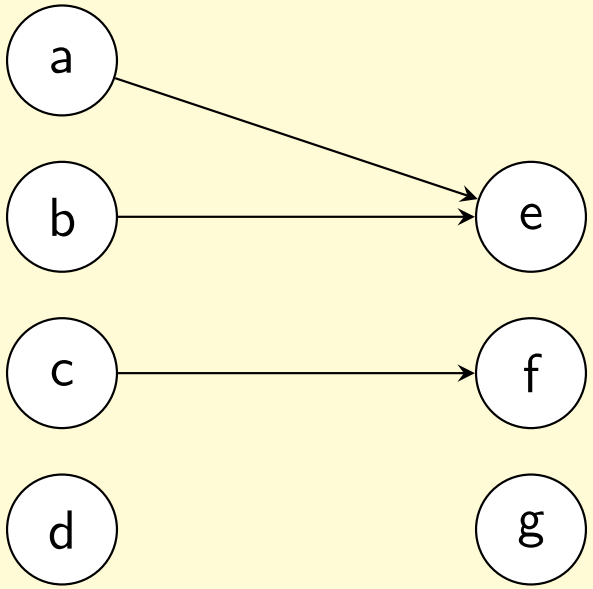
Is R a function? No. Element d gets mapped to both e and f.

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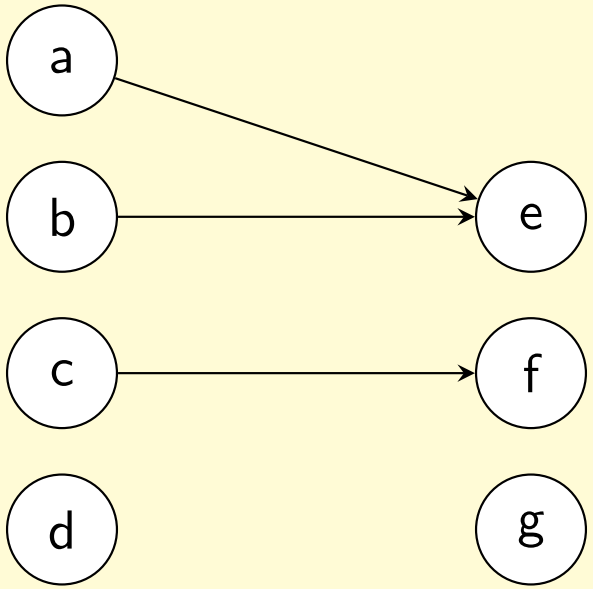
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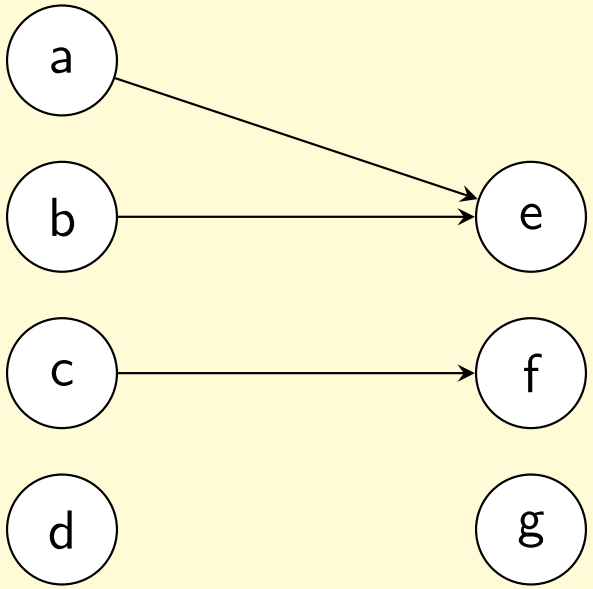
Is R a function? Not from A to B. Element d does not get mapped to anything.

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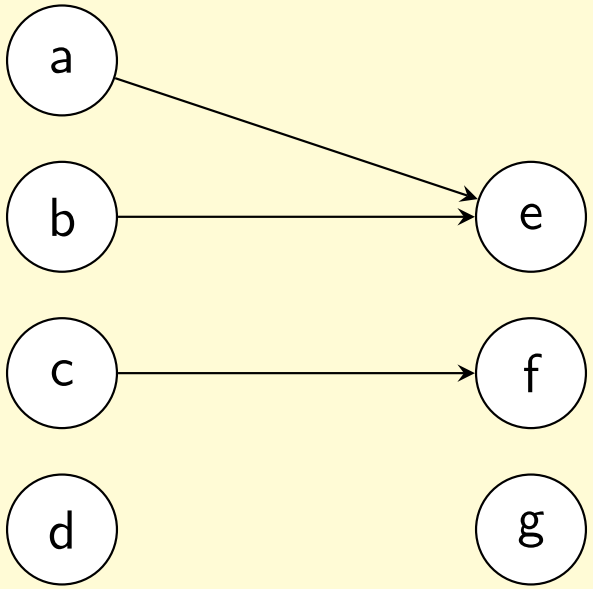
Is R a function? Not from A to B. Element d does not get mapped to anything. Is there a way to make it a function?

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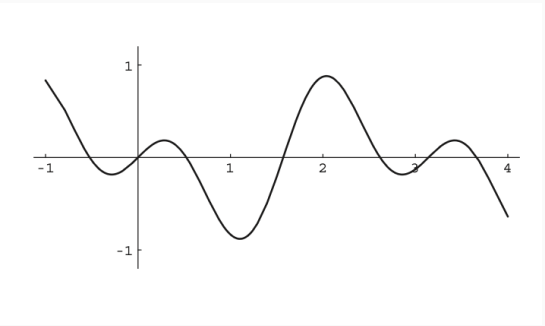
Let $A = \{a, b, c, d\}$ and $B = \{e, f, g\}$. Consider a relation R depicted below.



Is R a function? Not from A to B. Element d does not get mapped to anything. However, R is function from $\{a, b, c\}$ to B.

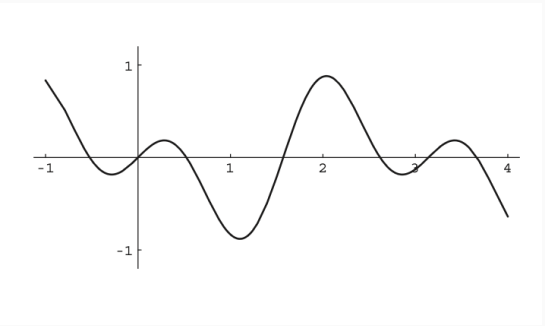
Motivation: Why are we doing this?

Often people think about functions in terms of graphs.



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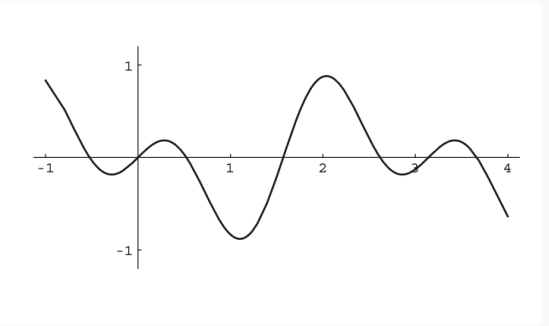


However... (Quote from Scheinerman pp. 170)

In discrete math, we are interested in functions to and from finite sets (or $\mathbb{N}$ or $\mathbb{Z}$). In such cases, traditional graphs of functions are either not helpful or nonsensical.

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Often people think about functions in terms of graphs.



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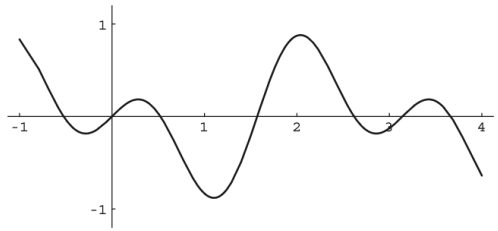
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Example

Let A be a finite set. We consider the function $f : 2^A \to \mathbb{N}$ defined by $f(x) = |x|$. *(Alert: the vertical bars in this context do not mean absolute value! To each subset of A, the function f assigns its size.)* There is no practical way to draw this as a graph on coordinate axes.

Motivation: Why are we doing this?

Often people think about functions in terms of graphs.



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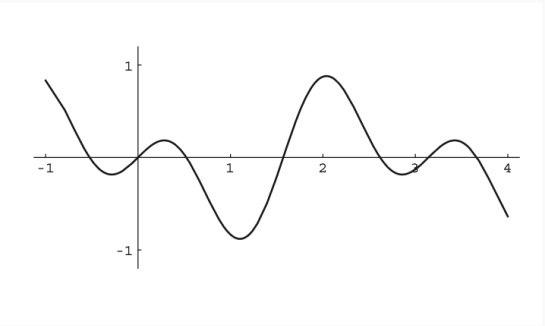
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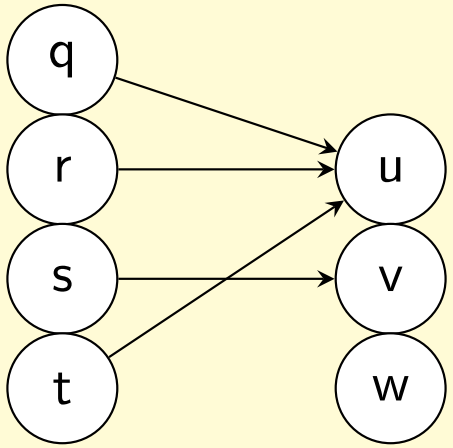
An application of the example. Let A be students in this class. Then 2^A is the set of all subsets of students in this class. f gives the number of students in each subset. How would you plot this? (Note: There are $|2^A| = 2^{51} \approx 1$ quadrillion subsets.)

Outline for Mini-Lecture:

- Overview of functions
- Domain, co-domain, and image
- Injectivity and surjectivity
- Inverse functions
- Applications (outside CSCI 246)

Poll

Let $A = \{q, r, s, t\}$ and $B = \{u, v, w\}$. Consider the function below.



What are the domain, codomain and image?

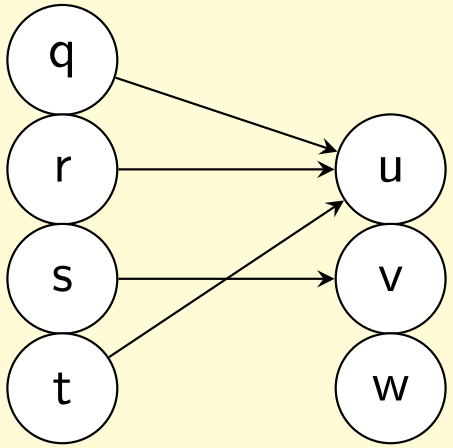
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The **image** (or **range**) of the function, in contrast, is the set of all elements that arise as $f(a)$ for some a in the domain. (It is the set of all elements in B that are actually "hit" by f).

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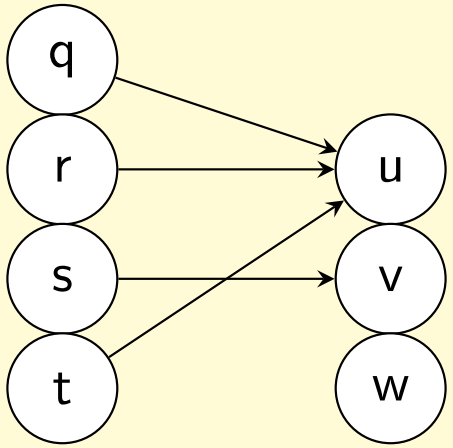
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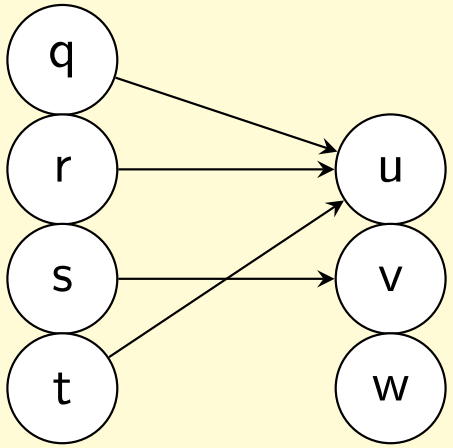
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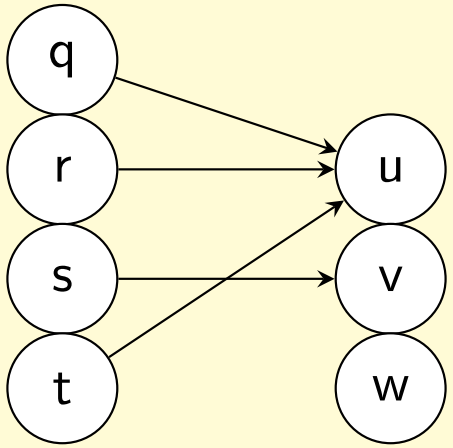
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What are the domain, codomain and image? The domain is $A = \{q, r, s, t\}$, the codomain is and $B = \{u, v, w\}$, and the image is $\{u, v\}$. Note that the image is smaller than the codomain B.

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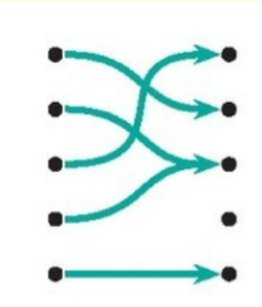
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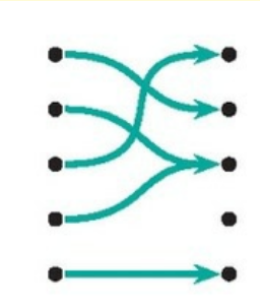
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Not injective
Not surjective

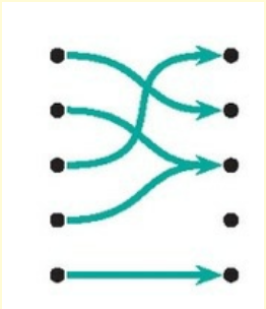
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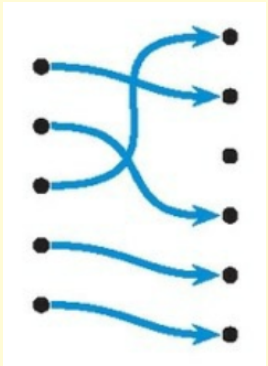
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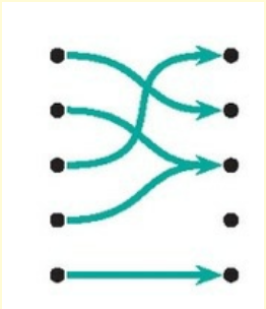
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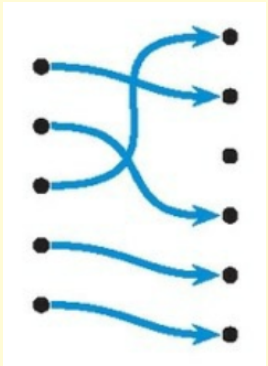
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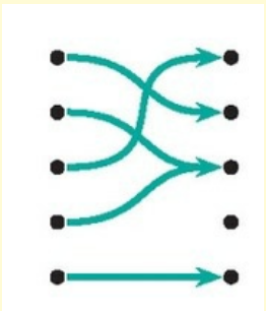
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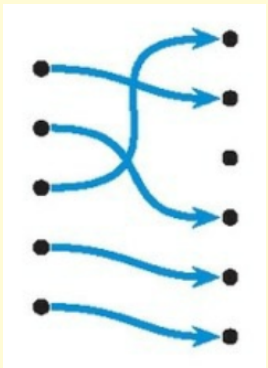
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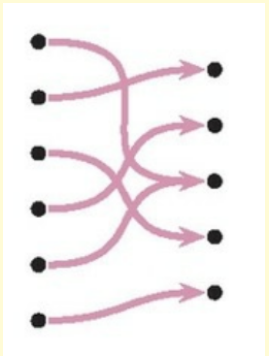
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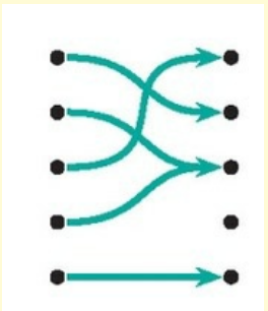
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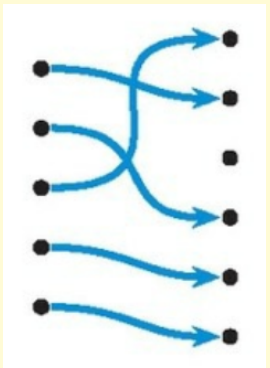
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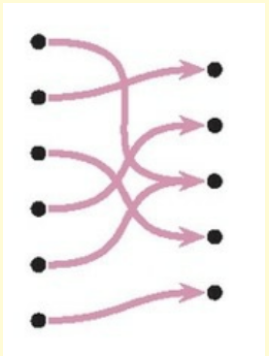
Poll



Not injective
Not surjective



Injective
Not surjective



Not injective
Surjective

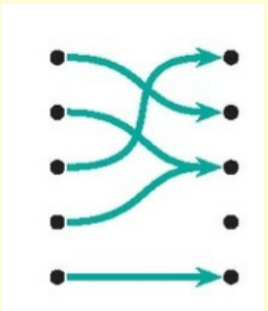
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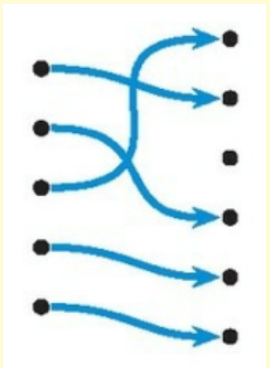
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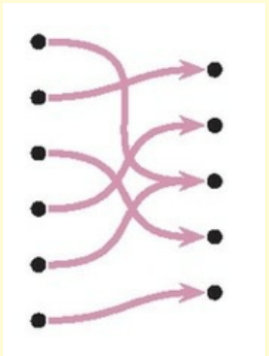
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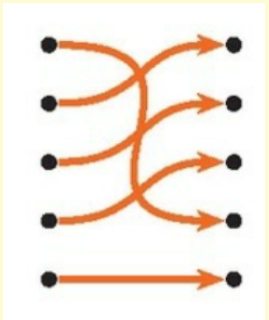
Not injective
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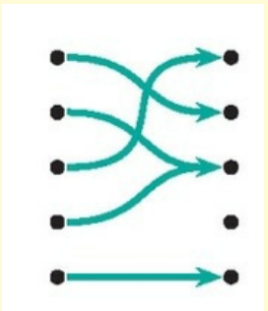
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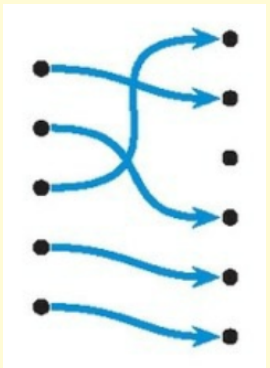
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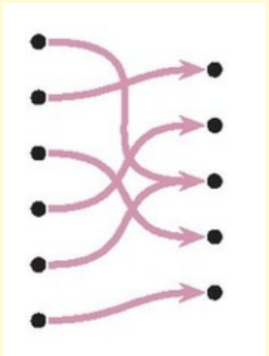
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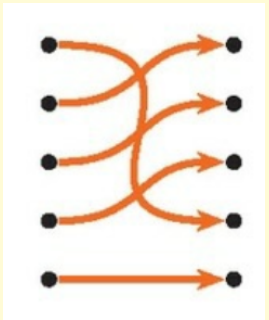
Not injective
Not surjective



Injective
Not surjective



Not injective
Surjective



Injective
Surjective

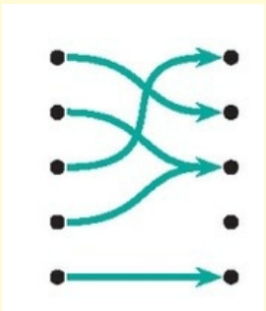
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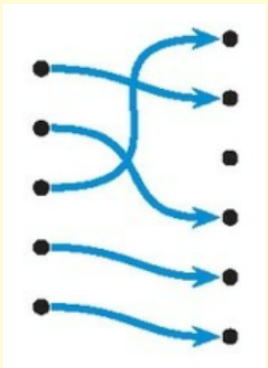
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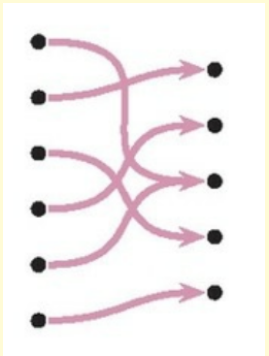
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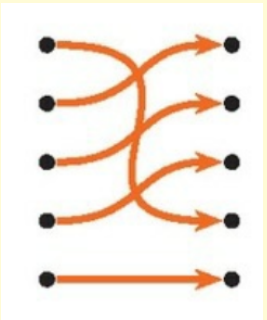
Not injective
Not surjective



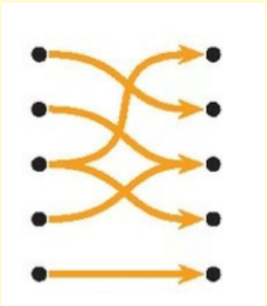
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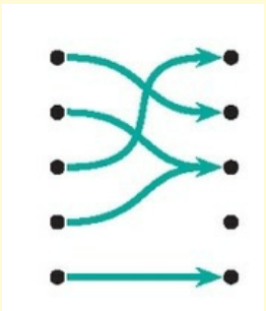
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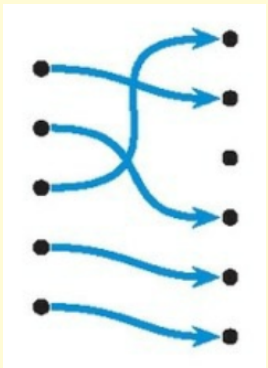
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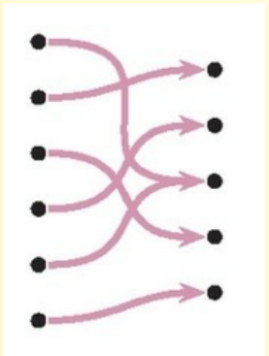
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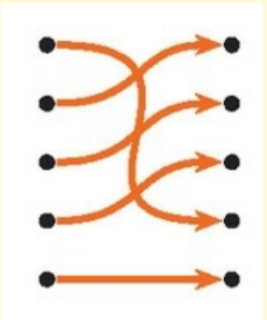
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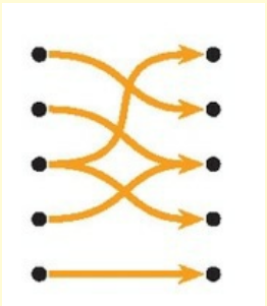
Injective
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Not injective
Surjective



Injective
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Not a function

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- Injectivity and surjectivity
- Inverse functions
- Applications (outside CSCI 246)

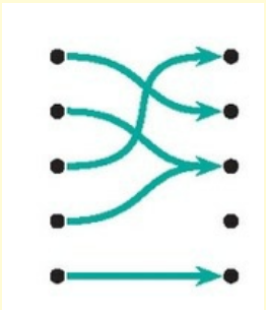
Definition (Scheinerman Def. 14.4)

Let R be a relation. The **inverse relation** of R, denoted R^{-1}, is the relation formed by reversing the order of all the ordered pairs of R.

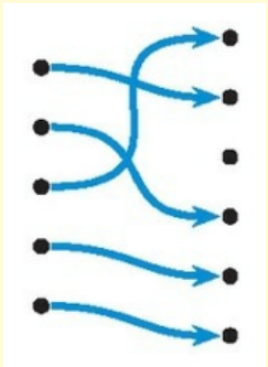
Theorem (Scheinerman Thm 24.21)

Let A and B be sets and let $f : A \to B$. The inverse relation f^{-1} is a *function* from B to A if and only if f is a bijection.

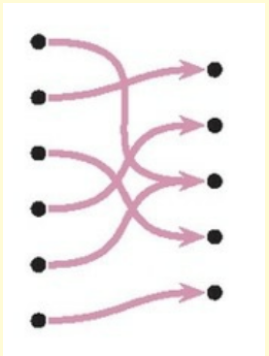
Poll



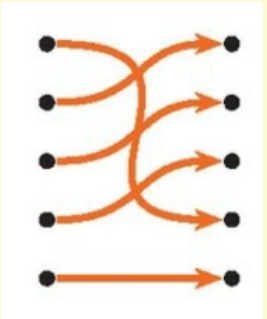
Not injective
Not surjective



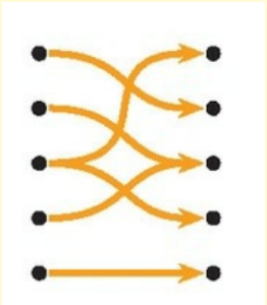
Injective
Not surjective



Not injective
Surjective



Injective
Surjective



Not a function

For which relations f above is f^{-1} a function? Why or why not?

Poll

Let $A = \{0, 1, 2, 3, 4\}$ and $B = \{5, 6, 7, 8, 9\}$.
Let $f : A \to B$ be defined by

$$f = \{(0, 5), (1, 7), (2, 8), (3, 9), (4, 7)\}$$

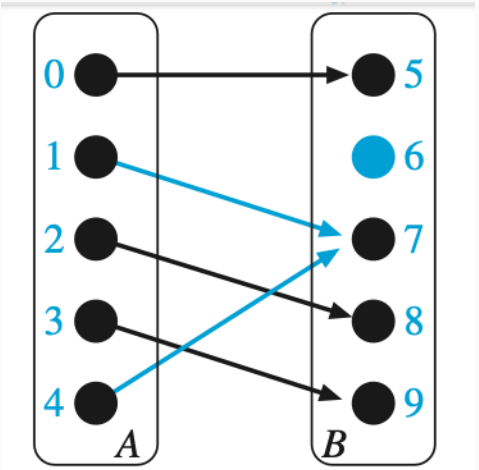
1. What is f^{-1}?
2. Is f^{-1} a function from B to A?

Poll

Let $A = \{0, 1, 2, 3, 4\}$ and $B = \{5, 6, 7, 8, 9\}$.
Let $f : A \to B$ be defined by

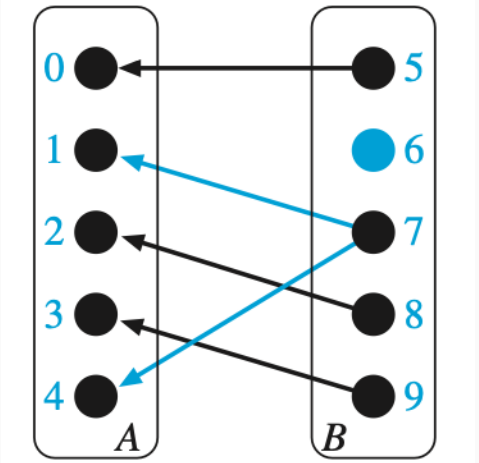
$$f = \{(0, 5), (1, 7), (2, 8), (3, 9), (4, 7)\}$$

1. What is f^{-1}?
2. Is f^{-1} a function from B to A?



Solution.

1. $f^{-1} = \{(5, 0), (7, 1), (8, 2), (9, 3), (7, 4)\}$.

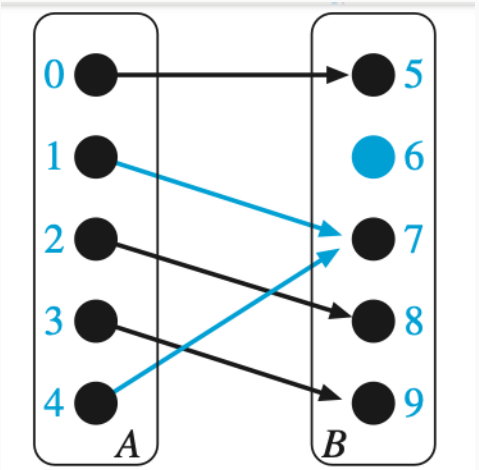


Poll

Let $A = \{0, 1, 2, 3, 4\}$ and $B = \{5, 6, 7, 8, 9\}$.
Let $f : A \to B$ be defined by

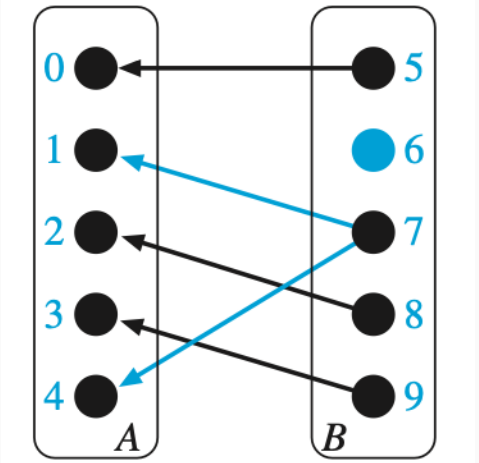
$$f = \{(0, 5), (1, 7), (2, 8), (3, 9), (4, 7)\}$$

1. What is f^{-1}?
2. Is f^{-1} a function from B to A?



Solution.

1. $f^{-1} = \{(5, 0), (7, 1), (8, 2), (9, 3), (7, 4)\}$.
2. No, f is not bijective. In particular:

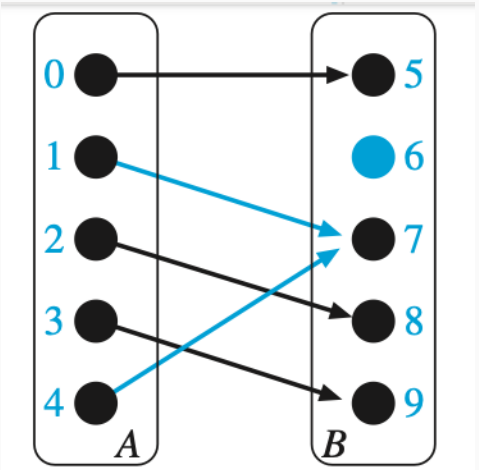


Poll

Let $A = \{0, 1, 2, 3, 4\}$ and $B = \{5, 6, 7, 8, 9\}$.
Let $f : A \to B$ be defined by

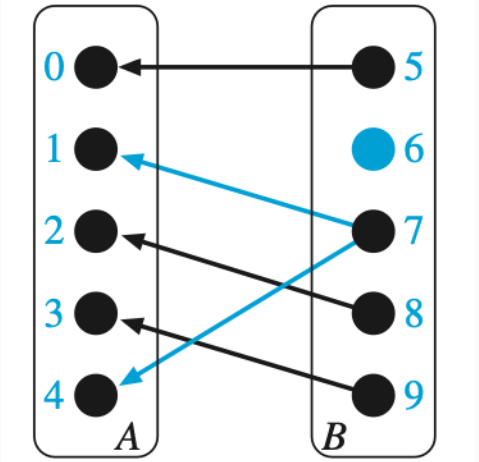
$$f = \{(0, 5), (1, 7), (2, 8), (3, 9), (4, 7)\}$$

1. What is f^{-1}?
2. Is f^{-1} a function from B to A?



Solution.

1. $f^{-1} = \{(5, 0), (7, 1), (8, 2), (9, 3), (7, 4)\}$.
2. No, f is not bijective. In particular:
   a. f is not surjective. So the input 6 doesn't map to *any* outputs.

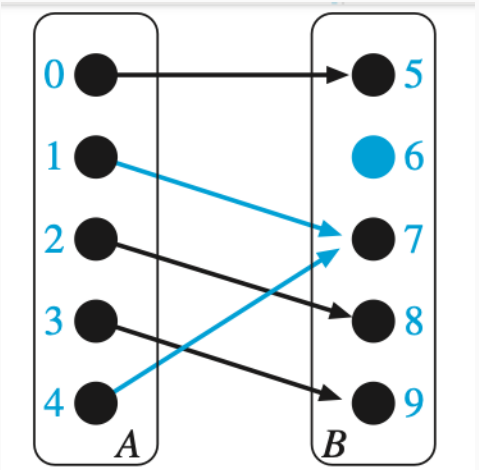


Poll

Let $A = \{0, 1, 2, 3, 4\}$ and $B = \{5, 6, 7, 8, 9\}$.
Let $f : A \to B$ be defined by

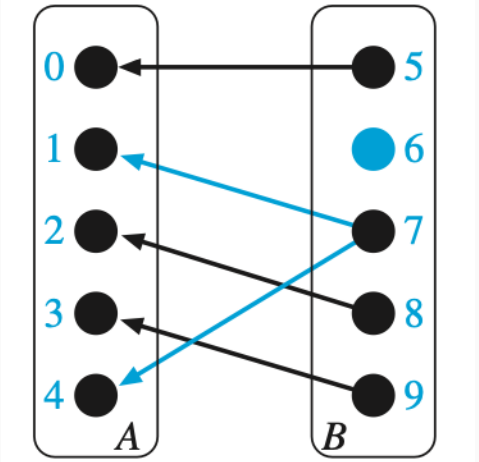
$$f = \{(0, 5), (1, 7), (2, 8), (3, 9), (4, 7)\}$$

1. What is f^{-1}?
2. Is f^{-1} a function from B to A?



Solution.

1. $f^{-1} = \{(5, 0), (7, 1), (8, 2), (9, 3), (7, 4)\}$.
2. No, f is not bijective. In particular:
   a. f is not surjective. So the input 6 doesn't map to *any* outputs.
   b. f is not injective. So the input 7 maps to *multiple* outputs.



Outline for Mini-Lecture:

- Overview of functions
- Domain, co-domain, and image
- Injectivity and surjectivity
- Inverse functions
- Applications (outside CSCI 246)

Spatiotemporal fields

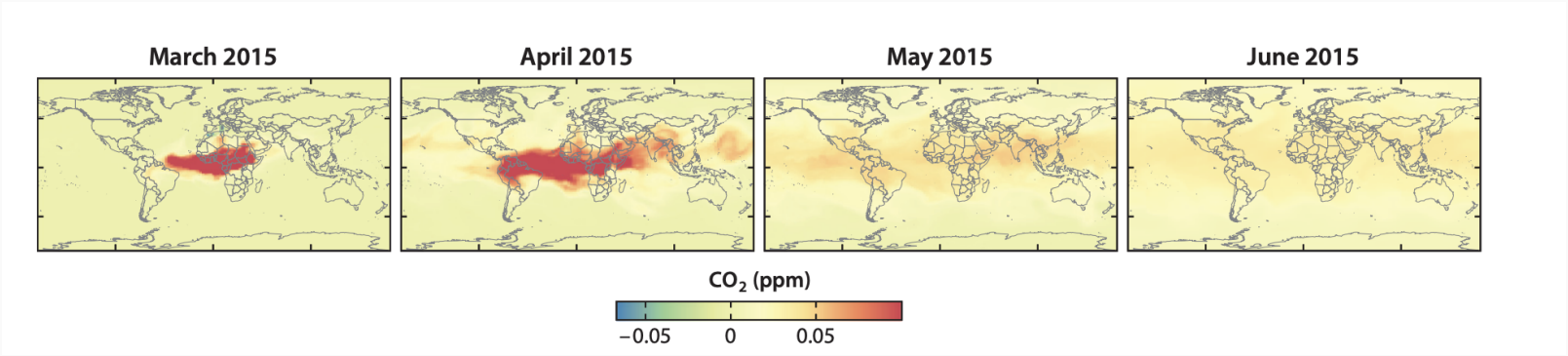


Figure 1: A spatiotemporal field showing CO2 changes, in units of parts per million (ppm), generated using the transport model MOZART.

Spatiotemporal fields

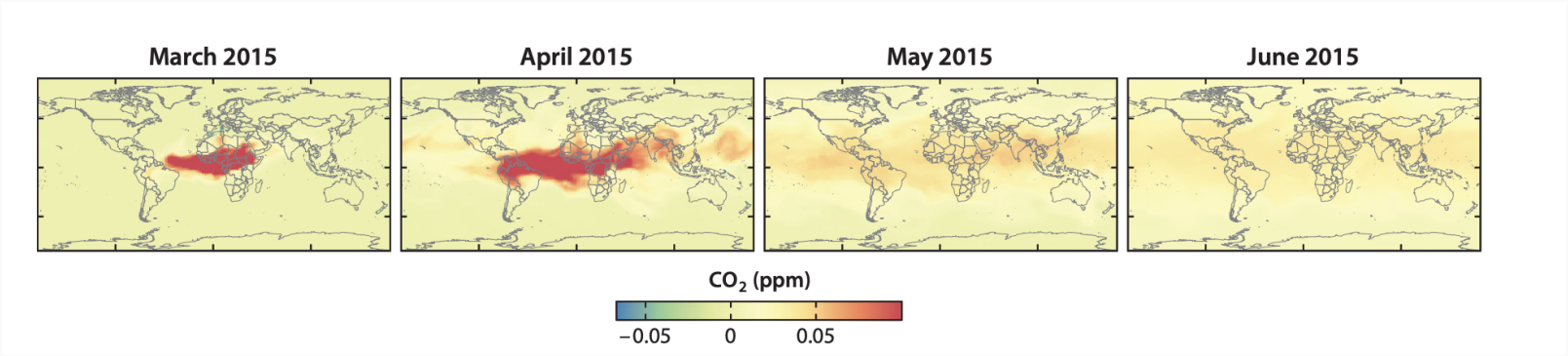
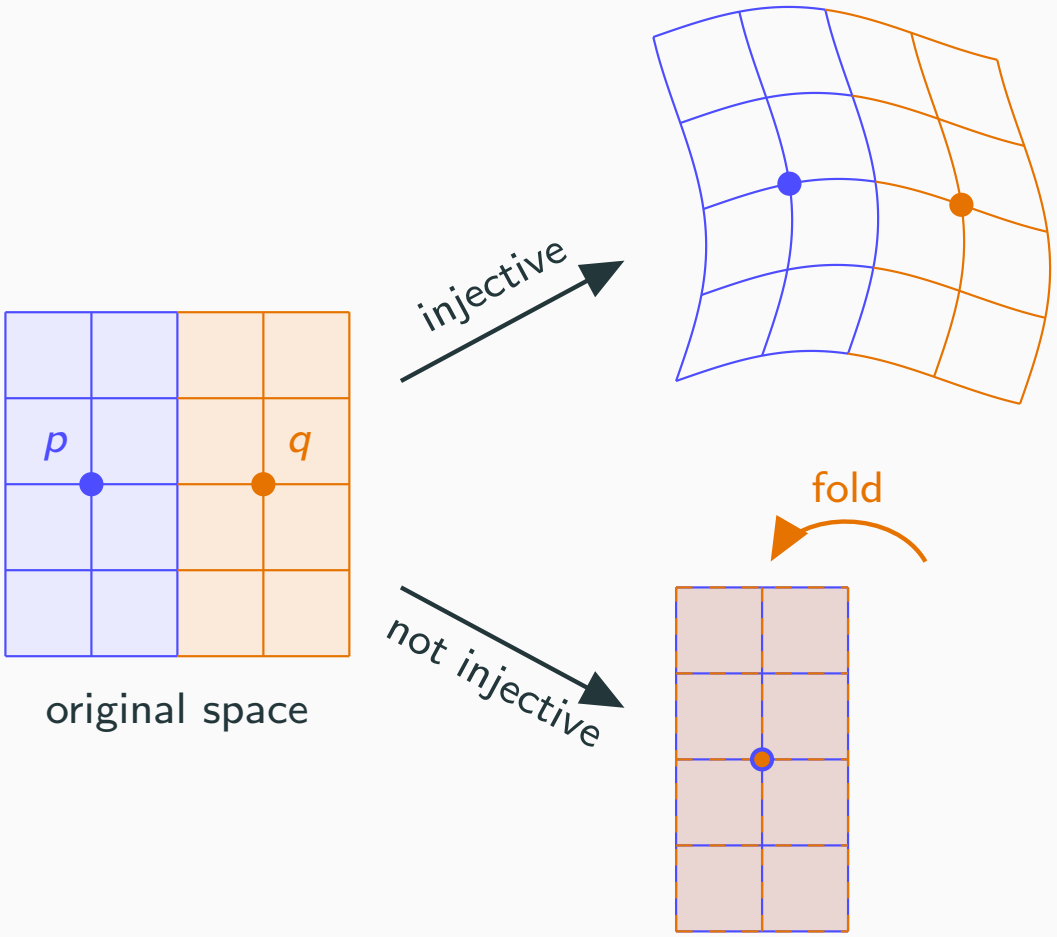


Figure 1: A spatiotemporal field showing CO2 changes, in units of parts per million (ppm), generated using the transport model MOZART.

"A straightforward way to model [complicated spatial fields] is first to warp the domain. [...] More recently, a class of machine-learning models has emerged [...] where the warping is carried out using a deep neural network. [...] [One approach] circumvents [pathologies] by forcing the mapping in each layer of the deep neural network to be injective, which precludes the possibility of space-folding, where space is mapped onto itself and can create undesirable spatial-prediction artifacts"

— Cressie et al. (2022), Basis-Function Models in Spatial Statistics.

Injective spatial maps



Space is stretched and bent, but **never overlaps**:

$p \neq q \;\Rightarrow\; f(p) \neq f(q)$

Space is **folded** onto itself:

$p \neq q$, but $f(p) = f(q)$

Group exercises

Random group assignments

Ashley, Isabelle: 10
Bailey, Tanna: 8
Blaisdell, Olyver: 6
Brown, Ashton: 11
Cardenas, Dominic: 11
Cutler, George: 1
Fabik, Roland: 7
Fellows, Audrey: 3
Fitzgerald, Ryan: 5
Fuller, Ryan: 7
Griffen, Clara: 1
Grutsch, Tanner: 10
Hager, Nathan: 3
Horner, Jayden: 12
Johnson, Jaret: 4
Kintzel, Samantha: 11
Lameres, Alexis: 2
Lee, Holden: 7
Mayala, Reece: 9
McLean, Connor: 3
Mcbride, Daniel: 12
Miller, Maverick: 4
Moss, Emily: 12
Mousseau, Matthew: 10
Neuman, Evan: 8
Putnam, John: 6
Ramos, Vydel: 5
Rathbun, Jedidiah: 9
Roberts, Matthew: 6
Roller, Nathan: 5
Ryan, Otis: 2
Shevtsov, Timothy: 1
Stensland, Emma: 2
Thiessen, Logan: 9
Thompson, Jaiden: 4
Vath, Ethan: 8

Group exercises

1. Let $A = \{1, 2, 3, 4\}$ and $B = \{5, 6, 7\}$. Let f be the relation
$$f = \{(1, 5), (2, 5), (3, 6), (?, ?)\}$$
Find replacements for (?,?) so that each of the following is true.
   a. The relation f is not a function from A to B.
   b. The relation f is a function from A to B but is not onto B.
   c. The relation f is a function from A to B and is onto B.
2. For each of the following relations, answer the following questions: (i) Is it a function? If not, explain why and stop. Otherwise, continue onward. (ii) What are its domain, codomain, and image? (iii) Is the function injective? (iv) Is it surjective?
   a) $\{(x, y) : x, y \in \mathbb{Z}, y = 2x\}$
   b) $\{(x, y) : x, y \in \mathbb{Z}, xy = 0\}$
   c) $\{(x, y) : x, y \in \mathbb{Z}, y = x^2\}$
3. Let $f : \mathbb{Z} \to \mathbb{N}$ by $f(x) = |x|$. (a) Is f one-to-one? (b) Is f onto?
4. Consider the "shifting" (or translation) function $f : \mathbb{Z} \to \mathbb{Z}$ given by $f(z) = z + k$ for some fixed $k \in \mathbb{Z}$. For instance $f(z) = z + 2$ shifts all integers two units to the right.
   a. Prove that f is a bijection.
   b. Express the inverse function f^{-1} with an explicit formula.

Solution to group exercise #1

Problem. Let $A = \{1, 2, 3, 4\}$ and $B = \{5, 6, 7\}$. Let f be the relation

$$f = \{(1, 5), (2, 5), (3, 6), (?, ?)\}$$

Find replacements for (?,?) so that each of the following is true.

a. The relation f is not a function from A to B.
b. The relation f is a function from A to B but is not onto B.
c. The relation f is a function from A to B and is onto B.

Solution.

a. The choice $(1, 6)$ means that $1 \in A$ is related to two different elements in B, which is not allowed. Moreover $4 \in A$ is not related to anything. Note that there are many possible other solutions. [In fact, any $(x, b) \notin \{(1, 5), (2, 5), (3, 6)\}$ where $x \in \{1, 2, 3\}$ and $b \in B$ will work.]
b. The choice $(4, 5)$ means that $7 \in B$ is not "hit" by f. Note that $(4, 6)$ would work as well.
c. $(4, 7)$.

Solution to Group Exercise #2

Problem. For each of the following relations, answer the following questions: (i) Is it a function? If not, explain why and stop. Otherwise, continue onward. (ii) What are its domain, codomain, and image? (iii) Is the function injective? (iv) Is it surjective?

a) $f_1 := \{(x, y) : x, y \in \mathbb{Z}, y = 2x\}$
b) $f_2 := \{(x, y) : x, y \in \mathbb{Z}, xy = 0\}$
c) $f_3 := \{(x, y) : x, y \in \mathbb{Z}, y = x^2\}$

Solution.

(i) Is it a function? f_1 and f_3 are functions because each x maps to exactly one y. f_2 is not a function because $x = 0$ maps to more than one (in fact, infinitely many) y's.

(ii) For all three functions, the domain and codomain are the integers. (This was explicitly stated in the problem.) The image for f_1 is the even integers. The image for f_3 is the perfect squares (i.e., $\{0, 1, 4, 9, \ldots\}$).

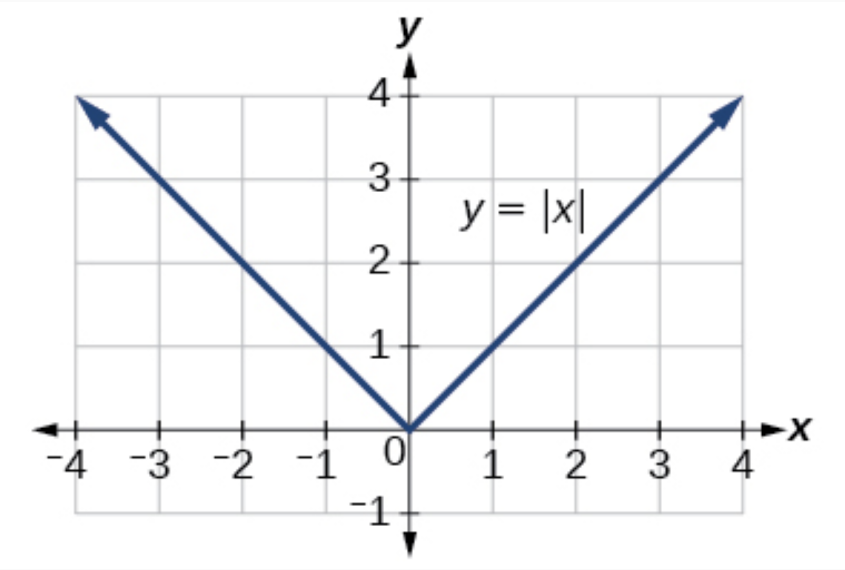
(iii) f_1 is injective, but f_2 is not. (Think: you can recover x from y for f_1, but not f_2).

(iv) f_1 is surjective, but f_2 is not. This is immediately clear from (ii), because a function is surjective if and only if its range equals its codomain. (Think: the realized function values covers the entire space of the codomain for f_1, but not f_2.)

Solution to group exercise #3

Problem. Let $f : \mathbb{Z} \to \mathbb{N}$ by $f(x) = |x|$. (a) Is f one-to-one? (b) Is f onto?

Solution.



a. No. For example, $f(1) = f(-1) = 1$. Hence, distinct inputs map to the same output.

b. Yes. Recall that $\mathbb{N} := \{0, 1, 2, 3, \ldots\}$. We can see from the plot that all of $\mathbb{N}$ is "hit" by f. For a more formal argument, we need to show that for each $n \in \mathbb{N}$, there is a $z \in \mathbb{Z}$ such that $f(z) = n$. This is easily satisfied by taking $z = n$ (or, for that matter $z = -n$).

Solution to group exercise #4

Problem. Consider the "shifting" (or translation) function $f : \mathbb{Z} \to \mathbb{Z}$ given by $f(z) = z + k$ for some fixed $k \in \mathbb{Z}$. For instance $f(z) = z + 2$ shifts all integers two units to the right. (a) Prove that f is a bijection. (b) Express the inverse function f^{-1} with an explicit formula.

Solution.

a. To show that f is a bijection, we need to show that it is injective and surjective.
   - Injective. By definition, a function f is injective if $a \neq a'$ implies $f(a) \neq f(a')$. We verify this statement by contraposition. That is, we prove the logically equivalent statement that $f(a) = f(a')$ implies $a = a'$. Now
   $$f(a) = f(a') \implies a + k = a' + k \implies a = a'.$$
   - Surjective. By definition, a function f is surjective if for any b in the codomain of the function, there is an a in the domain such that $f(a) = b$. Now note that we can obtain any $b \in \mathbb{Z}$ as $b = f(a)$ by setting $a = b - k$.

b. We can define f^{-1} by the mapping $f^{-1}(z) = z - k$.

Remark: Proof by contraposition

Earlier in the course, we showed that $A \implies B$ is logically equivalent to $(\texttt{not } B) \implies (\texttt{not } A)$. (For example, this can be shown by truth tables.) The transformed proposition is called the contrapositive. It may be easier to prove the contrapositive form of the statement, as we did above.

Bonus group exercises

Bonus group exercise

Consider the "zigzag" function $f : \mathbb{N} \to \mathbb{Z}$ given by

$$f(n) = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even} \\ -\frac{n+1}{2}, & \text{if } n \text{ is odd} \end{cases}$$

This function "zigzags" the integers like this

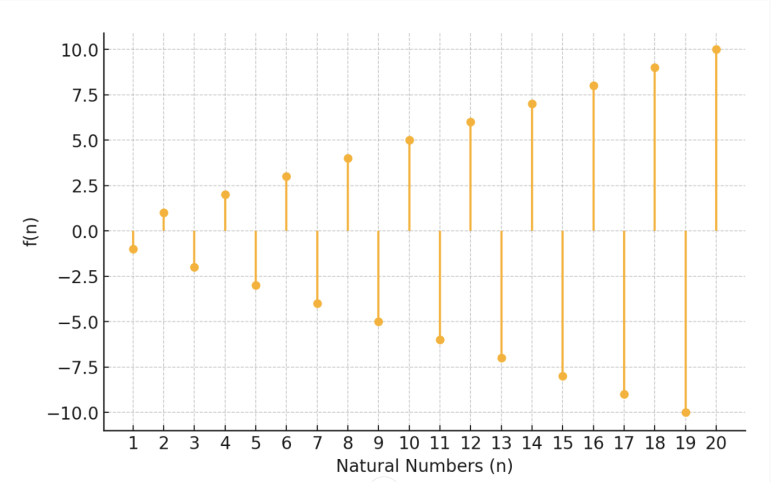
$$0 \to 0, 1 \to -1, 2 \to 1, 3 \to -2, \ldots$$

Prove that f is a bijection.

Solution (Part I) to bonus group exercise

Problem. Consider the "zigzag" function $f : \mathbb{N} \to \mathbb{Z}$ given by

$$f(n) = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even} \\ -\frac{n+1}{2}, & \text{if } n \text{ is odd} \end{cases}$$



This function "zigzags" the integers like this

$$0 \to 0, 1 \to -1, 2 \to 1, 3 \to -2, \ldots$$

Prove that f is a bijection.

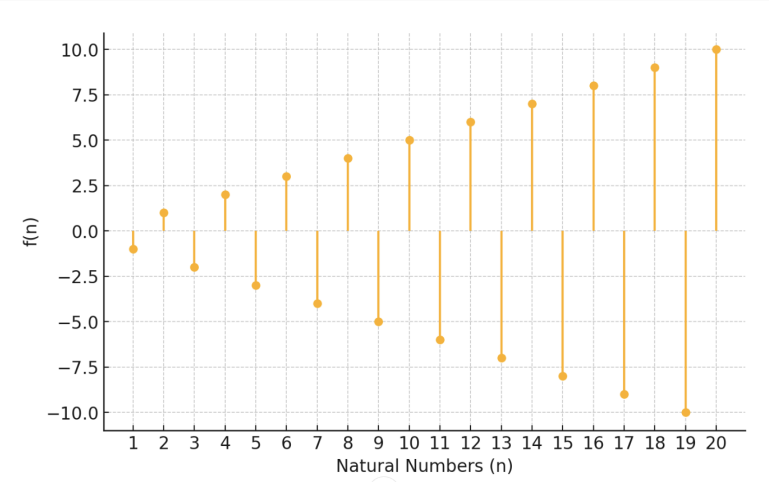
Solution. To show that f is a bijection, we need to show that it is injective and surjective.

- Injective. As with exercise #3a, to prove that f is injective, we show that $f(n) = f(n')$ implies $n = n'$. First note that if $f(n) = f(n')$, then n and n' must have the same parity, or else $f(n)$ and $f(n')$ would have different signs. Now
  - Suppose n and n' are both even. Then
  $$f(n) = f(n') \implies \frac{n}{2} = \frac{n'}{2} \implies n = n'.$$
  - Suppose n and n' are both odd. Then
  $$f(n) = f(n') \implies -\frac{n+1}{2} = -\frac{n'+1}{2} \implies n = n'.$$
- Surjective. See next page.

Solution (Part II) to bonus group exercise

Problem. Consider the "zigzag" function $f : \mathbb{N} \to \mathbb{Z}$ given by

$$f(n) = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even} \\ -\frac{n+1}{2}, & \text{if } n \text{ is odd} \end{cases}$$



This function "zigzags" the integers like this

$$0 \to 0, 1 \to -1, 2 \to 1, 3 \to -2, \ldots$$

Prove that f is a bijection.

Solution. To show that f is a bijection, we need to show that it is injective and surjective.

- Injective. See previous page.
- Surjective. By definition, a function f is surjective if for any $z \in \mathbb{Z}$, there is an $n \in \mathbb{N}$ such that $f(n) = z$. We proceed by cases. If $z = 0$, then we take $n = 0$. If $z > 0$, then we take $n = 2z$. If $z < 0$, then we take $n = -2z - 1$.

Remark: The strangeness of infinity

Sets with infinitely many members have some strange properties. If $f : A \to B$ where A and B are finite, then f being a bijection requires $|A| = |B|$. Intuitively, each member in A can't have exactly one "buddy" in B unless A and B are the same size. Here, however, we see that f gives a bijection between $\mathbb{N}$ and $\mathbb{Z}$, even though $\mathbb{N}$ is "smaller" than $\mathbb{Z}$ (in the sense that $\mathbb{N} \subsetneq \mathbb{Z}$).